

Quantitative Reasoning

Statics

Mathematics

Statistics

↓
Data

→ primary data / Raw

→ Secondary data

Statics

Defination

"It refers to the Science
Comprising methods which we
are uses in collection,
Presentation, Analysis and
Interpretation of numerical
data."

population

"The total number of observation which we are concerned"

Types

- Finite population
- INFinite population

Finite population

"The population whose numbers of element are limited or countable are called Finite population."

Example

the cars in a town.
the books in a library.

Infinite population

"The population whose number of elements are unlimited or not countable are called Infinite population."

Example

The number of the germs in the patient body is uncountable.

Sample

"The part of the population which represent the whole population."

Census / Complex Enumeration

"If the information is collected from each

element of the population
is called census.

parameter

"The numerical quantity
calculated from
population is called
parameter."

Statistic

"The numerical quantity
calculated from sample
is called statistic."

Type

→ Descriptive Statistics

→ Inferential Statistics

Descriptive Statistics

"The branch of Statistics that deals with presentation, collection, and analysis of data."

Inferential Statistics

"The branch of Statistics that deals with drawing inference about population on the basis of sample."

variable

"A characteristics which can vary from one individual to another."

Qualitative variable

"A variable which
can not be measured
numerically."

Example

Honesty, smile

Quantitative variable

"A variable which
can be measured
numerically."

Example

Age, weight, Height

Types of Quantitative variable

- Discrete variable
- Continuous variable

Discrete variable

"A variable which can not take each and every value with its interval."

Example

price

Continuous variable

"A variable which can take each and every value with its interval is called continuous variable."

Example

weight, height

Discrete variable / Jumping variable

"The Discrete variable is also called Jumping variable."

Data

"The Collection of Rows fact and figures."

Qualitative data

"The Data which can be describe by Qualitative variable."

Quantitative data

"The Data which can be describe by Quantitative variable."

Continuous Data

"The data which can be describe by Continuous variable."

Discrete Data

"The data which can be describe by discrete variable."

Application of Statistics:

The Application of Statistics is written as:

- Business management and industry
- Economics
- Biology and Medicines
- Research

Characteristics of Statistic

The character of Statistic is written as

- Statics are numerically expressed
- Statistics are aggregate of facts
- Statics for a pre-determined purpose.
- Statics are capable of being placed in relation to each other.

Measurement scale

• Nominal scale

Ranking scale / ordinal scale

Interval scale

Ratio Scale

Nominal scale

The classification or group of the observation into mutually exclusive qualitative categories or classes is said to constitute a nominal scale.

Example

Students are classified into male and female.

Ranking scale / ordinal scale

It includes the characteristics of a nominal scale and in addition has the property of ordering or ranking of measurement.

Example

performance of students
is rated as good, Excellent,
Fair etc.

Interval Scale

A measurement scale
passing a constant
interval size but
not a zero point.

Example

Temperature

Ratio Scale

It is a special
kind of interval
scale where the
scale of measurement
has a true zero
point.

Example

The ratio scale is used to measure weights, volumes, length etc.

Chapter no 02

Presentation of Data

Classification

"The process of arranging data into different classes or categories according to some common characteristics present in data."

- Qualitative Classification
- Quantitative Classification
- Geographical Classification
- Temporal/Chronological Classification

Qualitative Classification

when data are
classified according
to qualitative
characteristics.

Quantitative Classification

when data are
classified according
to quantitative
characteristics.

Geographical Classification

when data are
classified according
to geographical
characteristics.

Temporal classification

when data are
classified according
to temporal
characteristics

Types

- one way classification
- two way classification
- three way classification
- N-way classification

one way classification

A one-way classification
is a method of

categorizing data into
distinct groups or
classes based on a
single attribute or
variable.

Example
classifying Students by
gender (male/female)

Two way Classification

A two way classification
is a method of
categorizing data into
distinct groups or classes
based on two independent
variables or attributes.

Example

Classifying Students by gender
(male/female) and academic performance
(high/Low)

Three way Classification

A three way Classification
is a method of categorizing
data into distinct groups

or classes based on
three independent variables
on attributes

PUACP

presentation of Data

Classification

"The process of arranging data into different classes or categories according to some common characteristics present in data."

Chapter No 02

presentation of data

Classification

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Qualitative Classification

Quantitative Classification

Geographical Classification

Temporal/Chronological Classification

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Types

- one way classification
- two way classification
- three way classification
- N-way classification

one way classification

A one-way classification
is a method of
categorizing data into
distinct groups or
classes based on a
single attribute or
variable.

Examples

classifying students by
genders (male/female)

Two way classification

Defination

Y:

"A two way classification is a method of categorizing data into distinct groups or classes based on two independent variables or attributes."

Example

Classifying students by gender (male/female) and academic performance (high/low).

Three way classification

Definition

"A three way classification is a method of categorizing data into distinct groups or classes based on three independent variables or attributes."

Examples

Classifying students by
gender (male/female) & academic
performance (high/low) & ^{and} extracurricular
activities (yes/no).

N-way ClassificationDefinition

"N-way Classification also
known as multi-way classification
is a method of categorizing
data into distinct groups or classes
based on multiple independent variables
or attributes."

TableDefinition

Example: Classifying
customers by
demographics (age, location,
income)
behaviour and preferences

"The process of arranging
data into rows and
columns."

Table Tabulation

Box head

Stack

body of table

Source note

Frequency

Defination

6) The number of values specified in the class.

Class interval

Defination

"distance between two lower limit"

DAY: _____

DATE: _____

FrequencyTable

Class Limits Classes	Frequency (f)	Class Boundaries	Midpoint class mark	Calculated Frequency	Relative Frequency	Cumulative Frequency
5-9	3	4.5-9.5	7	3	3/34	3/34
10-14	4	9.5-14.5	12	7	4/34	7/34
15-19	7	14.5-19.5	17	14	7/34	14/34
20-24	12	19.5-24.5	22	26	12/34	26/34
25-29	6	24.5-29.5	27	32	6/34	32/34
30-34	2	29.5-34.5	32	34	2/34	34/34

DATE: _____

Table

Table is a form of visual data that summarize data into row or column.

Tobulation

Tobulation is a method of presenting numeric data in rows and columns in a logical and systematic manner.

Classes

A grouping of values by which data is binned for compilation of a frequency distribution.

Frequency

The number of times a particular value for data item has been observed to occur.

DAY: _____

DATE: _____

Class Boundaries

Class boundaries are the numbers used to separate classes. It is located upper limit of class and lower limit of classes.

Mid point/Class Mark

The average value or the mid point value of a given interval (is x) in frequency distribution.

Commulative Frequency

It is defined as the total of frequencies distributed over different classes.

Relative Frequency

The ratio of number of times a value of data occurs in set of all outcomes total number of outcomes.

Commulative relative frequency

The proportion of times a particular value is observed in class intervals.

Class interval

Difference between two successive lower limits is said to be class interval.

Foot Note

The foot note are given at the bottom of the table. It is used to explain the contents of table.

Source Note

Source note are given at the end of table which indicate the compiling agency & publication of data.

Prefatory Note

The prefatory note is given after the title of Table. It is used to explain the contents of data.

Box-head

Consists of the Column head, Stub head and Stamber head that describe

DATE: _____
column is columns to
which they refer.

Stub

The heading for
different row is Caption row
& this part of row
caption is called Stub.

Body of table

The main part of
table in which numerical
figures are put.

Frequency distribution

is representation
either in or tabular form
that displays number of
observation, with in given interval.

DAY: _____

DATE: _____

Cumulative Frequency

It is defined as relative frequency.

Range of data

The difference b/w the maximum value and minimum value.

$$\text{Range} = \text{max} - \text{min}$$

Q.1

Make a Frequency distribution?

$$\text{Range} = \text{max} - \text{min}$$

$$\text{no. of classes} = C = 1 + 3.3 \log(N)$$

$$\text{Class interval} = h = \frac{R}{C}$$

74, 65, 90, 77, 66, 40, 62, 104, 96, 103, 96, 96

83, 95, 82, 84, 57, 90, 65, 62, 60, 18, 85, 58, 57

52, 105, 66, 114, 88, 116, 100, 120, 106, 91, 80, 121

96, 83, 54, 92, 72, 99, 75, 56, 52, 55, 64, 64

76 99 6 63 6 110 6 84 6 104 6 95 6 97 6 82 6 78 6 78

Range = Max - Min

$$R = 121 - 49$$

$$R = 72$$

$$\text{No. of classes} = C = 1 + 3.3 \log(N)$$

$$= C = 1 + 3.3 \log(60)$$

$$= 6.86$$

$$C \approx 7$$

$$h = \frac{R}{C} = \frac{72}{7} = 10.28$$

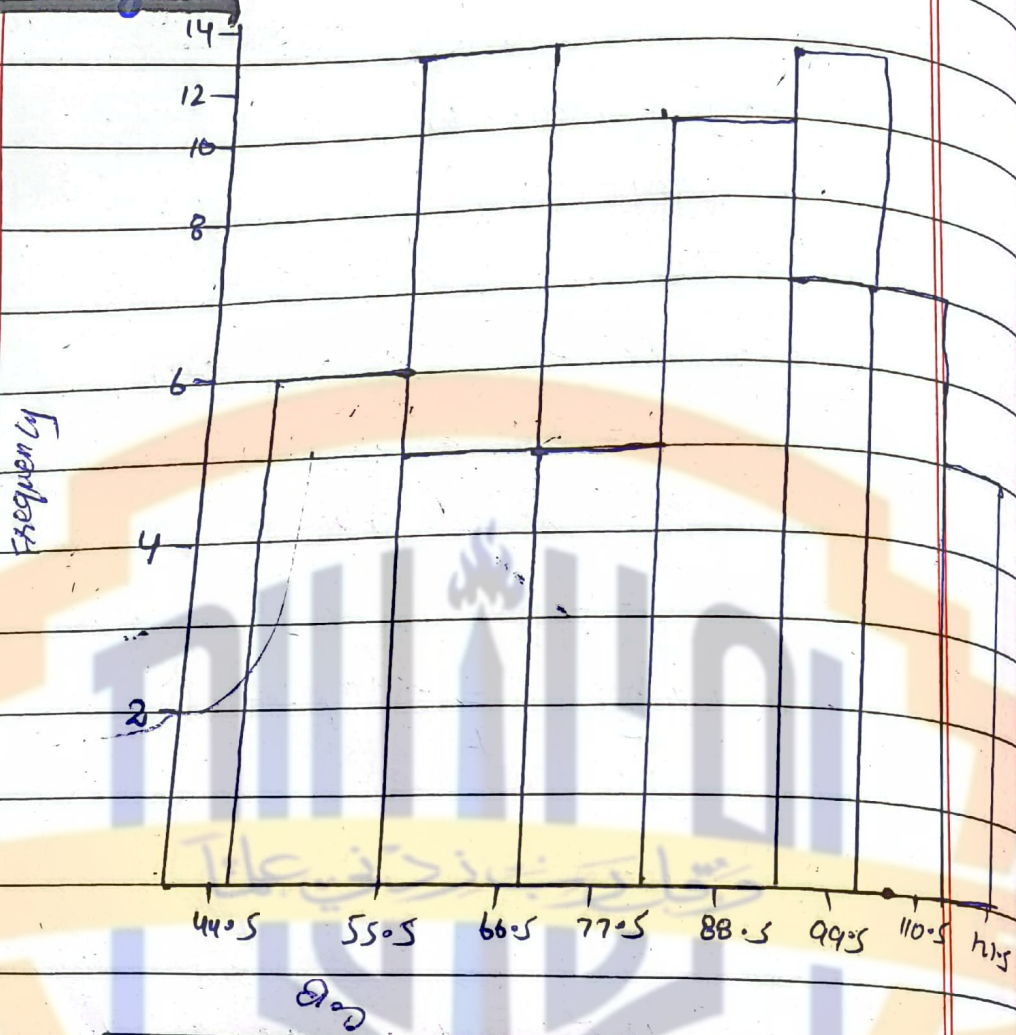
$$h \approx 11$$

classes	Tally	Freq frequency	C.B	Midpoint	C.F	S.F	Coef
45-55		6	44.5-55.5	50	6	6/60	6/60
56-66		13	55.5-66.5	61	19	13/60	19/60
67-77		5	66.5-77.5	72	24	5/60	24/60
78-88		11	77.5-88.5	83	35	11/60	35/60
89-99		13	88.5-99.5	94	48	13/60	48/60
100-110		7	99.5-110.5	105	55	7/60	55/60
111-121		5	110.5-121.5	116	60	5/60	60/60

DAY: _____

DATE: _____

Histogram

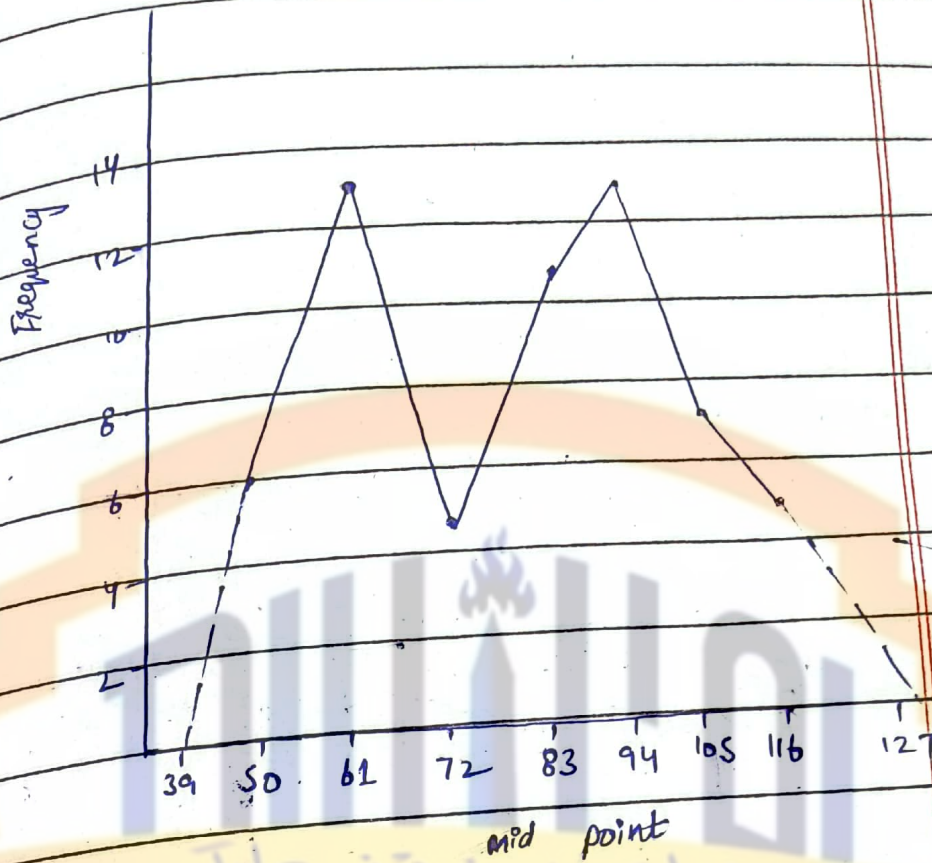


Frequency polygon

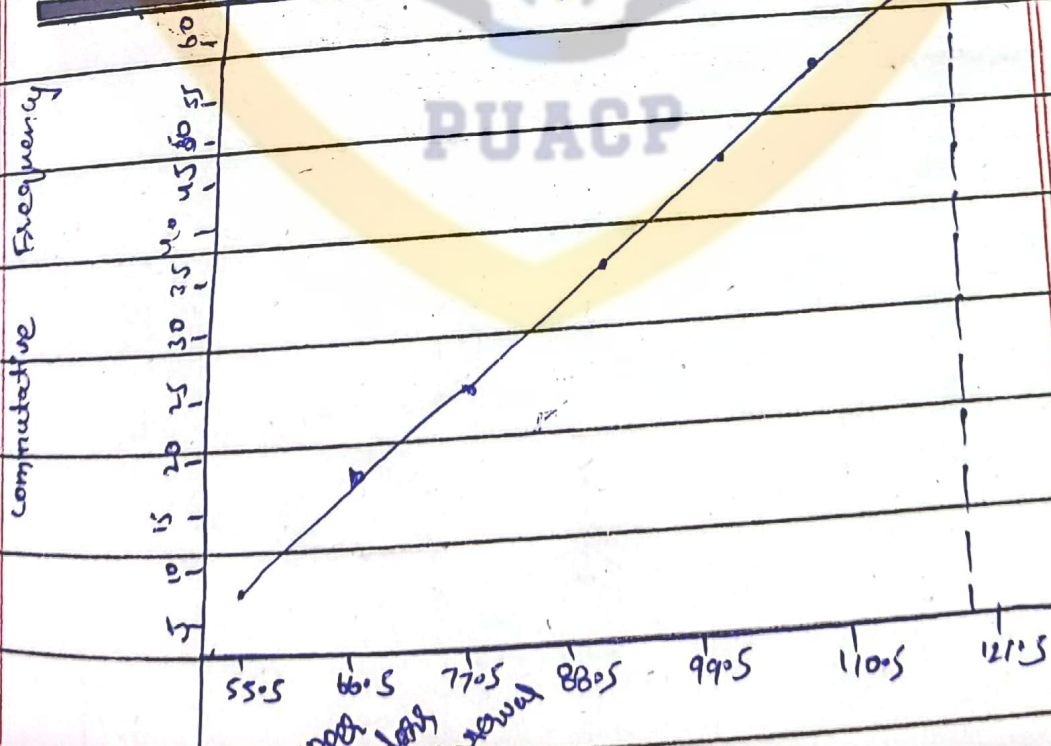
Diagram

DATE: _____

DAY: _____



Cumulative Frequency polygon



DAY: _____

DATE: _____

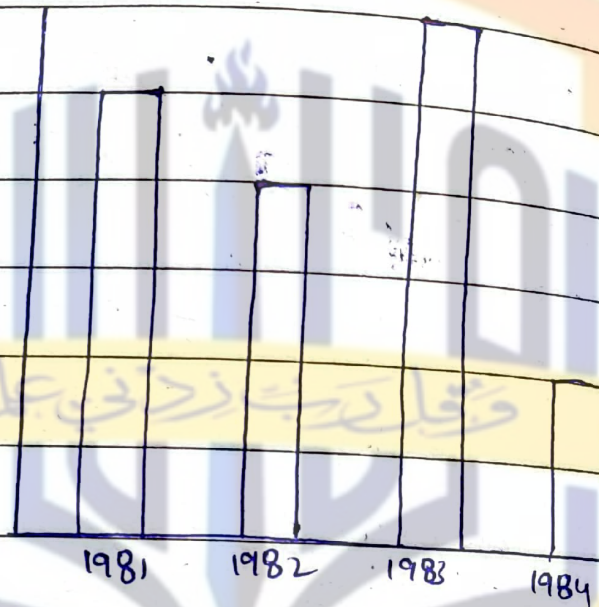
Simple Bar chart

Multiple Bar chart

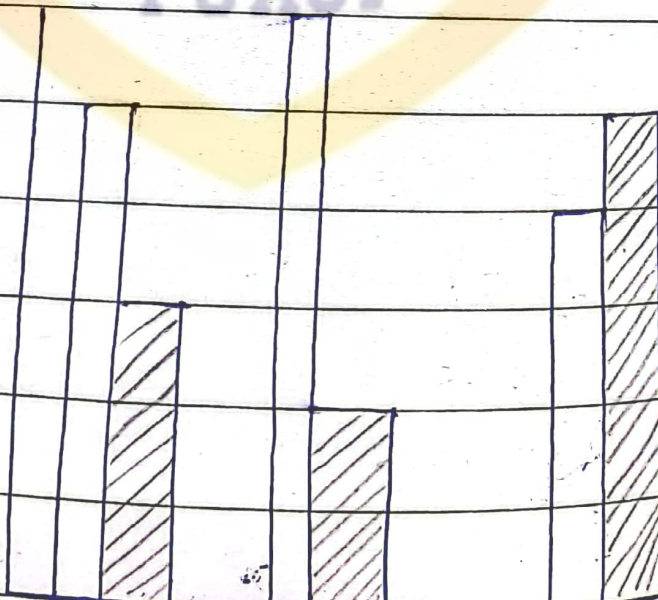
Component Bar chart

pie chart

Simple Bar chart

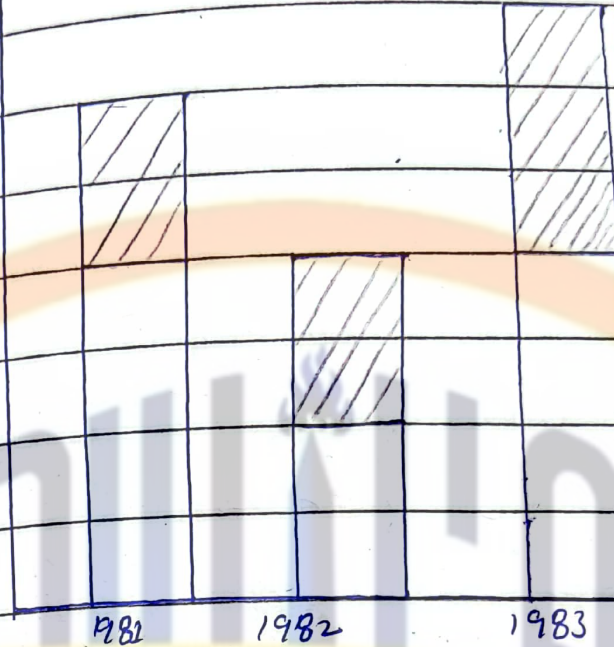


Multiple Bar chart



DATE: _____

Component Bar



QND19 photo on mobile
QND20 photo on mobile

QND:06

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06
09
13
19

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